Part C: Naming & Formula of Multivalent Ionic Compounds

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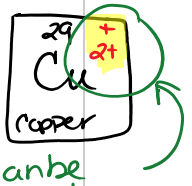
PART C: NAMING & FORMULA OF MULTIVALENT IONIC COMPOUNDS

multiple different ion charges



can be Cu^{+} ← copper(I) ion
Cu^{2+} copper(II) ion

Have a look at your periodic table, can you find any elements that have more than one ion charge? List an example below:

$Cu^{+}, Cu^{2+}, Co^{2+}, Co^{3+}, Ti^{3+}, Ti^{4+}, Fe^{2+}, Fe^{3+}$

Multivalent metals are metals that can form more than one type of ion. For example, copper can form an ion with a +1 or a +2 charge.

That means that copper can give up either 1 or 2 electrons when forming an ionic compound with a non-metal ion.

1. FROM COMPOUND NAME --> CHEMICAL FORMULA

When we name multivalent metals we need a way of distinguishing what ion charge of the multivalent metals is involved.

For this we use Roman Numerals

- Ni^{2+} or nickel (II) ion called "nickel two" and shows the nickel ion has an ion charge of 2+.
- Ni^{3+} or nickel (III) ion is called "nickel three" and shows the nickel ion has an ion charge of 3+.

Table 4.7
Roman Numerals

Metal Ion Charge	Roman Numeral
1+	I
2+	II
3+	III
4+	IV
5+	V
6+	VI
7+	VII

Cr^{3+} Cl^{-}

For example, the name chromium (III) chloride tells you that that chromium ion in the compound is 3^{+} not Cr^{2+} ion ← could be either bc chromium is multivalent... but the (III) tells us which

When we write the chemical formula for an ionic compound containing a multivalent metal, the same steps we have already learned still apply.

swap + drop.

You can use the charge balancing method or the criss cross method.

The only thing that is new is that we will see a Roman Numeral in the name of the compound telling us which ion charge is involved.

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Writing Formulas of Compounds Containing a Multivalent Metal

Steps	Examples	
	manganese(IV) sulfide	cobalt(III) oxide
1. Identify each ion and its charge.	Mn^{4+} S^{2-}	Co^{+3} O^{-2}
2. Determine the total charges needed to balance positive with negative.	2 : 4 (swap!) $(+4)+(+4)+(-2)+(-2)+(-2)+(-2)=0$	2 : 3 (can't simplify) $(+3)+(+3)+(-2)+(-2)+(-2)=0$
3. Note the ratio of positive ions to negative ions.	1 : 2 (simplify ÷2) $(+4)+(-2)+(-2)=0$	
4. Use subscripts to write the formula. A "1" is not shown in the subscripts.	MnS_2 (drop!) = MnS_2 (don't have to write 1)	Co_2O_3 ("little numbers" subscripts.)

PRACTICE

(a) copper(I) nitride

Listicns: Cu^{+1} N^{3-}

SWAP : 3 : 1 (don't need to simplify)

Check Charges: $(+1)+(+1)+(+1)+(-3)=0$

Drop : Cu_3N_1 = Cu_3N

(b) iron(II) phosphide

Fe^{2+} P^{3-} (swap!)

3 : 2

$(+2)+(+2)+(+2)+(-3)+(-3)=0$ ✓ balanced.

Chemical Formula ⟹ Fe_3P_2 (drop!)

(c) chromium (III) chloride

Cr^{3+} Cl^{-1}

1 : 3

$(+3)+(-1)+(-1)+(-1)=0$ balanced ✓

$CrCl_3$

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HW April 15

ASSIGNMENT #4: Writing formulas of Ionic Compounds with MULITVALENT ions

This assignment is to be completed below in the space provided.

Practice Problems

1. **Write the formulas** of the following compounds containing multivalent metals.

(a) manganese(II) oxide MnO

(b) manganese(IV) oxide MnO_2

(c) chromium(II) bromide $CrBr_2$

(d) chromium(III) bromide $CrBr_3$

(e) lead(IV) chloride $PbCl_4$

(f) iron(III) phosphide FeP

(g) tin(II) sulfide SnS

(h) tin(II) nitride Sn_3N_2

(i) mercury(II) fluoride HgF_2

(j) tin(IV) nitride Sn_3N_4

(k) copper(I) iodide CuI

(l) copper(II) iodide CuI_2

(m) Iron (II) iodide FeI_2

(n) Nickel (III) sulfide Ni_2S_3

(o) Iron (II) fluoride FeF_2

(p) Copper (II) chloride $CuCl_2$

(q) Tin (IV) oxide SnO_2

(r) Titanium (IV) oxide TiO_2

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2. FROM CHEMICAL FORMULA --> COMPOUND NAME

Naming an Ionic Compound with a Multivalent Metal

The compound Fe_2O_3 is the main source of iron in the making of steel, which is used for things like cutlery to shipping freighters.

Pure Fe_2O_3 is reddish in colour and is used as a pigment in some paints.

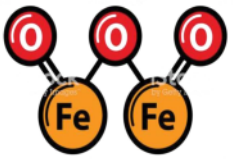
Fe is multivalent Fe^{2+} or Fe^{3+}

What is the name of Fe_2O_3?

Step 1: Identify the ions. Fe^{2+} or Fe^{3+} + O^{2-}

Step 2: "Reverse Swap + drop"

Work backwards using either charge balancing or criss cross methods to determine which iron ion is involved in the compound.



Fe^{2+} or Fe^{3+} O^{2-}

② reverse the "swap"

2 : 3

① reverse "drop"

Fe_2O_3

Step 3: $(+3)+(+3)+(-2)+(-2)+(-2)=0$ balance charges.

Write the name of the compound using a Roman Numeral to indicate the charge of the multivalent metal ion.

Don't forget you still need to name the non-metal with a "-ide" ending!

ions used: Fe^{3+} O^{2-}

iron (III) oxide

NAMING Ionic Compounds that Contain a Multivalent Metals: *(reverse "swap and drop")*

Naming Ionic Compounds Containing a Multivalent Metal

Steps	Examples ① Au_3N	② PdS_2
1. Identify the metal.	Au = gold.	Pd = Palladium
2. Verify that it can form more than one kind of ion by checking the periodic table.	Au^+ or Au^{3+}	Pd^{2+} or Pd^{4+}
3. Determine the ratio of the ions in the formula.	Au_3N_1	4:2 → 2:1 → 1:2 (S^{2-})
4. Note the charge of the negative ion from the periodic table.	N is always N^{-3}	Pd_1S_2
5. The positive and negative charges must balance out. Determine what the charge needs to be on the metal ion to balance the negative ion.	Au^+ Au^+ Au^+ N^{-3} (+1)+(+1)+(+1)+(-3)≠0	Pd^{4+} S^{2-} S^{2-} (+4)+(-2)+(-2)=0 ✓
6. Write the name of the compound. Au^+ N^{3-}	gold (I) nitride	Palladium (IV) sulfide

This ratio has been simplified. work backwards and ×2

② reverse swap.
① reverse drop

PRACTICE Each of these compounds contains a multivalent metal ion. That means that the name of the metal ion will contain a Roman numeral, which you will need to determine. Write the names of the following compounds.

(a) PbF_4 — Lead (IV) fluoride
(b) FeI_2 — iron (II) iodide
(c) HgI_2 — mercury (II) iodide
(d) Hg_3N_2 — mercury (III) nitride
(e) ~~Fe_2O_3~~ Fe_2O_3 — iron (III) oxide
(f) Sn_3P_4 — tin (IV) phosphide
(g) MnS — manganese (II) sulfide → Mn_1S_1 (Mn^{2+} S^{2-}) (+2)+(-2)=0
(h) MnS_2 — manganese (IV) sulphide
(i) VCl_5 — vanadium (V) chloride
(j) Ni_2S_3 — nickel (III) sulphide.

check non-metal first.

Ni^{2+} Ni^{3+} → Ni^{3+} Ni^{3+} S^{2-} S^{2-} S^{2-} (+6)+(-6)=0

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Homework

ASSIGNMENT #5: Naming Ionic Compounds with MULITVALENT ions
This assignment is to be completed below in the space provided.

HW April 15th (Assignment #4)

Example: Write the names of the following:

for multivalent ions, the name MUST indicate what the ⊕ ion charge is.

- $FeCl_3$ (Fe^{3+}) — iron (III) chloride
- $CuCl$ (Cu^+) — copper (I) chloride
- SnF_4 (Sn^{4+}) — tin (IV) fluoride
- $PbCl_2$ (Pb^{2+}) — lead (II) chloride
- Fe_2S_3 (Fe^{3+}) — iron (III) sulfide

1. Write the **names** of the compounds with the following ions:
   a. Co^{3+} and O^{2-} — cobalt (III) oxide
   b. Cu^+ and Br^- — copper (I) bromide
   c. Cu^{2+} and Cl^- — copper (II) chloride
   d. Mn^{4+} and S^{2-} — manganese (IV) sulfide

c. Cu^{2+} and Cl^- copper (II) chloride

d. Mn^{4+} and S^{2-} manganese (IV) sulfide

2. Write the **names** of the following compounds. Each contains an ion of a multivalent metal.

a. FeO iron (II) oxide

b. Cu_3N copper (I) nitride

c. SnS_2 $Sn^{+2}S^{-1}$ $Sn^{+4}S^{-2}$ tin (IV) sulfide

d. Sn_3N_2 tin (II) nitride

e. Ni_2S_3 nickel (III) sulfide

f. $MoCl_3$ molybdenum (III) chloride

g. PbF_4 lead (IV) fluoride

h. TiS_2 titanium (IV) sulfide

3. Write the **chemical formula** for the following compounds.

a. lead (IV) chloride $PbCl_4$

b. lead (II) sulfide PbS